🔑 **Answer Key — Teacher Copy · Lesson 2-1**

MSTAR PRACTICE WORKSHEET

Lesson 2-1: Understand Statistical Questions

Grade 6 Mathematics · Standard 6.DS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which of the following is a statistical question?

- **A** How many points did each player score this season?
- **B** How many points are scored for a free throw?
- **C** What sport does this team play?
- **D** How many halves are in a soccer game?

> "How many points did each player score?" is statistical because different players will have different scores — there is variability in the data.

Question 2 SELECTED RESPONSE — CORRECT: C

Which question is NOT a statistical question?

- **A** How many home runs did each player hit this year?
- **B** How many miles does each runner train per week?
- **C** How many innings are in a baseball game?
- **D** How tall is each player on the basketball team?

> "How many innings are in a baseball game?" has one fixed answer (9) — no variability. The other questions all produce different answers for different people.

Question 3 SELECTED RESPONSE — CORRECT: A

Why is 'How many hours of sleep did each student get last night?' a statistical question?

- A Because different students got different amounts of sleep
- B Because it asks about sleep
- C Because hours are numbers
- D Because the answer is always 8 hours

> A statistical question anticipates variability — different students slept different amounts, so the data will vary.

Question 4 SELECTED RESPONSE — CORRECT: C

A coach asks: "How many hours does each athlete on the track team sleep the night before a meet?" Why is this a statistical question?

- A Because it is about sports
- B Because it has a numerical answer
- C Because the answers will vary from athlete to athlete
- D Because the coach asked it

> A statistical question anticipates variability. Different athletes sleep different amounts, so the data collected will vary.

Question 5 SELECTED RESPONSE — CORRECT: D

Which of these is a statistical question?

- A What time is it right now?
- B What is my shoe size?
- C What is 7×8?
- D How tall are the students in my class?

> A statistical question anticipates variability — students' heights differ, so the data will vary.

Question 6 SELECTED RESPONSE — CORRECT: A

Which of these would collect data that VARIES (making it a statistical question)?

- **A** How many siblings does each student in the class have?
- **B** How many days are in one week?
- **C** How many minutes are in one hour?
- **D** What is 12 + 12?

> Number of siblings varies from student to student, so the collected data will vary — that makes it statistical. The others each have one fixed answer.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Which question is a STATISTICAL question?

- (A) How many minutes did I sleep last night?
- (B) How many students are in our class?
- (C) What time does school start?
- (D) How many minutes did each student in our class sleep last night?

> A statistical question anticipates answers that VARY. Different students slept different amounts, so the data will vary. The other three each have exactly one fixed answer.

- **A:** This asks about one person, so it has exactly one answer. A statistical question anticipates answers that vary across a group.
- **B:** A class has one size, so this has a single answer. Statistical questions expect the answers to differ.
- **C:** School starts at one fixed time, so there is nothing to vary. Ask something that would give different answers from different students.

Part B — correct answer: C

What makes the other three questions non-statistical?

- (A) They are about school instead of about students.
- (B) They use smaller numbers.
- (C) Each has a single fixed answer, so there is no variability to describe.
- (D) They cannot be answered with data.

> Variability is the whole point. A question with one fixed answer produces one value, and there is no distribution to summarize.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D

Select ALL questions that are statistical questions:

- A How tall is each player on the team?
- B How many hours per week do sixth graders spend reading?
- C What is my shoe size?
- D How long does it take students in our class to get to school?
- E What is the capital of Maryland?

> A, B, and D each anticipate answers that vary across people. C has one fixed answer about one person, and E has one fixed factual answer.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Write one statistical question and one non-statistical question about your school's cafeteria. Explain why each question is or is not statistical, and describe what the data would look like for the statistical question.

> **Model answer:** A strong statistical cafeteria question asks about each person, such as 'How many minutes does each student wait in the lunch line?' — the data would be a spread of values like 2, 5, 7, and 11 minutes. A non-statistical one, such as 'How many tables are in the cafeteria?', has a single fixed answer, so there is nothing to collect or compare.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
